PECO312C : Machine Learning :Concepts and Applications Laboratory

Teaching Scheme	Examination Scheme	
Practical: 2 Hrs./Week	CIA:	20 Marks
	ESE:	30 Marks
Credits: 1	Total:	50 Marks

Prerequisite Course: IT for Engineers, Python Programming

Course Objectives

1. **To impart practical skills in applying core machine learning algorithms** such as regression, classification, clustering, and recommendation systems using real-world datasets across various domains (e.g., retail, banking, telecom, agriculture).
2. **To develop the ability to preprocess, clean, and visualize data effectively**, enabling students to identify patterns, trends, and anomalies essential for building reliable ML models.
3. **To enable students to evaluate and deploy machine learning models**, ensuring model efficiency, fairness, and interpretability, with exposure to concepts like REST API deployment and TinyML for edge computing.

Course Outcomes (COs):

After successful completion of the course, student will be able to

	Course Outcome (s)	Bloom's Taxonomy Level	Bloom's Taxonomy Descriptor
CO1	**Apply appropriate machine learning techniques** (e.g., linear regression, logistic regression, SVM, decision trees, Naive Bayes, K-NN) to solve domain-specific problems and assess performance using standard metrics like accuracy, precision, recall, F1-score, and ROC-AUC.	3	Apply
CO2	**Perform end-to-end machine learning workflows**, including data cleaning, feature engineering, model building, hyper parameter tuning, and visualization of results for interpretation.	3	Apply
CO3	Design and implement machine learning-based solutions using modern tools and frameworks suited for industry deployment	3	Apply

Mapping of Course Outcomes to Program Outcomes (POs) & Program Specific Outcomes (PSOs):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	3	-	-	-	-	-	-	-	-	3	-	-
CO2	3	3	3	2	2	-	-	1	-	-	-	3	-	-
CO3	3	3	3	3	3	-	-	-	-	-	-	3	2	-

Course Contents

Student Guidelines for Practical Implementation and Evaluation

General Instructions

1. **Total Assignments:** You are provided with 11 practical assignments in the syllabus. You must complete and submit any 8 out of the 11.

2. **Platform Requirement:** You must use any of the following platforms for implementation:

- Anaconda
- Jupyter Notebook
- VS Code
- PyCharm
- Google Colab

3. Dataset Requirement:

Use standard real-world datasets (e.g., MNIST, UCI datasets, Kaggle datasets) relevant to the problem definition of each assignment.

Datasets must be preprocessed and properly cited if taken from online sources.

Implementation and Submission Guidelines

4. Coding Standards:

Your code should be well-commented, modular, and readable.

Use appropriate libraries like pandas, numpy, matplotlib, seaborn, scikit-learn, TensorFlow, etc.

5. Execution Demonstration:

You must demonstrate each assignment implementation to the course teacher during lab hours or designated slots.

6. Write-up Submission:

Prepare a write-up for each practical assignment.

It should include:

i. Problem Statement

ii. Dataset Description

iii. Step-wise Algorithm/Flowchart

iv. Screenshots of Output

v. Result & Conclusion

Write-up must be handwritten neatly and clearly in a separate journal or practical book.

This write-up will be evaluated for marks, so maintain it well.

Evaluation Scheme (Per Practical - Total 10 Marks) - Implementation (Code Quality & Output), Performance (Understanding, Execution & Viva, Write-up (Neatness, Clarity, Completeness).

Term Work and Practical Exam

- Term Work: 25 Marks (Based on practical performance across 8 assignments)
- Practical Exam: 50 Marks (Final practical exam as per university schedule)

Best Practices

- Backup your work regularly on cloud platforms like Google Drive or GitHub.
- Do not plagiarize – copied work will be penalized.
- Be punctual and consistent in submissions.
- Focus on understanding algorithms, not just executing code.

	List of Assignments	No. of Hours	COs
1.	**Data Cleaning and Visualization for Retail Sales Analysis -** Given a retail sales dataset, perform data cleaning (handle missing values, normalize numerical features) and create visualizations (histograms, boxplots, correlation matrices) to explore seasonal patterns in sales	4	CO2
2.	**Linear Regression for House Price Prediction -** Build and evaluate a linear regression model to predict house prices based on size, location, number of rooms, and amenities. Evaluate the model.	2	CO1
3.	**Classification of Loan Applications Using Logistic Regression -** Train a logistic regression model to predict whether a loan application will be approved or not based on applicant information. Evaluate using confusion matrix, precision, recall, F1-score, and ROC-AUC.	4	CO1
4.	**Credit Risk Assessment in Banking (Model Evaluation Metrics) -** Develop a classification model to predict loan default using a real-world financial dataset. Evaluate the model using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC to environments.	2	CO1
5.	**Real-time Fraud Detection System (Model Deployment & Efficiency)** - Build and deploy a fraud detection model using logistic regression or XGBoost. Implement the model as a RESTful API and evaluate its inference time and memory usage under different loads.	2	CO3
6.	**Customer Churn Prediction using Decision Tree Classifier -** Develop a Decision Tree classifier to predict whether a telecom customer is likely to churn based on usage statistics (call duration, plan type, complaints, billing issues, etc.). Evaluate the model using accuracy, confusion matrix, and F1-score. Visualize the tree for interpretability.	2	CO1, CO2
7.	**Email Spam Detection using Naive Bayes Classifier -** Build a Naive Bayes classifier to classify emails as spam or not spam based on email text (subject, body content, etc.). Apply text preprocessing (tokenization, stop word removal) and TF-IDF vectorization.	2	CO1, CO2
8.	**Product Recommendation Category Classification using K-NN -** Use K-NN to classify products into categories (e.g., electronics, clothing, books) based on user ratings, reviews, and product metadata. Experiment with different values of K and distance metrics	2	CO1, CO2
9.	**Handwritten Digit Recognition using Support Vector Machine (SVM)** - Train an SVM classifier on the MNIST dataset to recognize handwritten digits (0-9). Perform hyperparameter tuning (kernel type, C value) and evaluate performance using precision and recall	2	CO1 CO2
10.	**Bias Detection and Fairness Evaluation in Hiring Algorithms** - Use a hiring dataset (e.g., UCI Adult Income dataset) to train a classification	2	CO1, CO3

	model for hiring decisions. Evaluate the model not only with traditional metrics (accuracy, F1-score, etc.) but also with fairness metrics (e.g., demographic parity, equal opportunity). Analyze the model's bias across gender and race groups and suggest mitigation strategies.		
11.	**Edge AI for Smart Agriculture using TinyML -** Implement a lightweight ML model (e.g., decision tree or quantized neural net) that can run on an edge device like Arduino or Raspberry Pi to detect crop disease or monitor soil moisture levels. Explore deployment using TensorFlow Lite and evaluate model performance under limited compute and memory conditions.	2	CO3

Text Books:

1. **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow** (3rd Edition) — *Aurélien Géron*
   - Covers data cleaning, model building, evaluation, hyperparameter tuning, deployment, TinyML (TensorFlow Lite).
2. **Pattern Recognition and Machine Learning** — *Christopher M. Bishop*
   - Strong foundation for classification algorithms (SVM, Naive Bayes, Logistic Regression), model evaluation, theory behind ML techniques.
3. **Introduction to Machine Learning with Python: A Guide for Data Scientists** — *Andreas C. Müller, Sarah Guido*
   - Focused on practical applications using Scikit-Learn; includes regression, classification, pipelines, model evaluation.

Reference Books:

1. **Machine Learning: A Probabilistic Perspective** — *Kevin P. Murphy*
   - In-depth coverage of probabilistic models, fairness metrics, and advanced evaluation strategies.
2. **TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers** — *Pete Warden & Daniel Situnayake*
   - Focuses on Edge AI and TinyML, directly relevant to your assignment on smart agriculture.
3. **Machine Learning Yearning** — *Andrew Ng* (available as free PDF)
   - Practical guidance on how to structure ML projects, choose models, evaluate, and deploy effectively.

eLearning Resources:

1. **Coursera — Machine Learning by Andrew Ng**
   Foundation in supervised learning, model evaluation, deployment principles — highly recommended starter.
2. **Fast.ai — Practical Deep Learning for Coders**
   - Hands-on projects, deployment, and fairness; excellent for modern ML practices.
3. **TensorFlow Lite and TinyML tutorials (TensorFlow official website + Edge Impulse Academy)**
   - Free, practical guides for deploying models on edge devices (e.g., Arduino, Raspberry Pi) with resource constraints